Multimedia Signal Processing - 2nd Module

January, 26 2022

1. **Question (8 pts)** Consider the scalar quantizer described by the following input output relation

$$y = q(x) = \begin{cases} y_1 & \text{if } x \leq -3, \\ y_2 & \text{if } -3 < x \leq 1, \\ y_3 & \text{if } 1 < x \leq 3, \\ y_4 & \text{if } 3 < x \leq 4, \\ y_5 & \text{if } x > 4. \end{cases}$$

The input of the quantizer is a continuous source X defined by the following uniform probability distribution function (pdf)

$$p_X(x) = \begin{cases} \frac{1}{8} & \text{if } -4 \leq x \leq 4, \\ 0 & \text{otherwise.} \end{cases}$$

Consider the discrete source Y observed at the output of the quantizer.

   (a) Determine the cardinality (i.e., the number of unique symbols) of the alphabet of Y.

   (b) Compute the probability mass function (pmf) of the source Y.

   (c) Compute the entropy $H(Y)$ of Y.

   (d) Design the Huffman code to encode symbols of Y.

   (e) Compare the average code length of the Huffman code to the length of the Shannon code and to $H(Y)$.

   (f) Recalling that the mean squared error (MSE) introduced by the quantizer in an interval defined by the endpoints a_k and a_{k+1} and reproduction level y_k is

$$\text{MSE}_k = \int_{a_k}^{a_{k+1}} (y_k - a)^2 p_X(a) da, \tag{1}$$

   define the values y_1, y_2, y_3 and y_4 that minimize the MSE introduced by the quantizer on X.

2. **Question (4 pts)** With the help of a block diagram, describe how an uncompressed RGB color image can be compressed with JPEG coding scheme. Highlight which steps take into account visual redundancy and / or visual irrelevancy, if any.

3. **Question (3 pts)** Explain with an example the concept of noise-masking-tone and tone-masking-noise in the frequency domain. Define the signal-to-mask ratio (SMR) explaining how this can prove useful in perceptual audio coding.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] It is possible for a quantizer to introduce zero distortion on a discrete source.
     TRUE: if I apply a quantizer to a discrete source obtained with the same quantizer, I am not introducing any more distortion
   - [T] [F] Lossless coding can be applied to sources characterized by an infinite-element alphabet.
     FALSE: by definition
   - [T] [F] A discrete source X with an alphabet composed by four symbols can have entropy $H(x) = 2$.
     TRUE: it's the maximum entropy that can be obtained in case the p.m.f is uniform
   - [T] [F] Any one-to-one code is also a uniquely decodable code.
     FALSE: uniquely decodable codes are a subset of one-to-one codes
   - [T] [F] In a video coding scheme it is possible to estimate motion vectors with a precision smaller than a single pixel.
     TRUE: see slides on half-pixel / quarter-pixel motion estimation
   - [T] [F] A close-loop DPCM scheme always provides less distortion than any PCM scheme at the same bitrate.
     FALSE: see slides on DPCM coding gain
   - [T] [F] In a speech coding scheme based on DPCM, it is good practice to tune the long-term residual on the pitch of the speech.
     TRUE: long-term predictor should be used to attenuate the effect of "peaks" present in a speech signal due to its pitch.
   - [T] [F] Video decoders have to know which motion vector search strategy has been used by the encoder to correctly decode frames.
     FALSE: the decoder needs the motion vectors written in the bitstream according to the used standard. No info about the way these vectors have been obtained is used at decoder side.

5. **Question (Matlab - 10pts)**

   Implement a KLT-based coding and decoding scheme for an audio signal by following these steps.

   Signal generation

   (a) Generate an auto-regressive process of order one (i.e., AR(1)) of 1000 samples. Consider the model $x(n) = \rho * x(n-1) + w(n)$, where $\rho = 0.99$ and $w(n)$ is extracted from a Gaussian distribution with zero mean and standard deviation equal to 2.

   (b) Clip the value of this signal in the range $[-15, 15]$ and round each sample to the closest integer. This is your input signal `x`.

   KLT encoding

   (a) Compute the 2×2 KLT matrix you would use to encode the signal `x` with a transform coding scheme with block size equal to 2 samples.

   (b) Compute the transform coefficients for each 2-samples block of `x`.

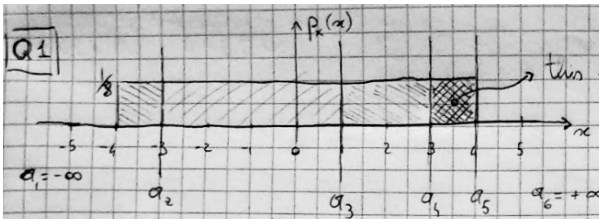
   (c) Quantize the coefficients with a uniform scalar quantizer with quantization step equal to 1.

   KLT decoding

(a) Decode the quantized coefficient by inverting KLT transform coding.

(b) Compute the introduced MSE.

Q1



a) x is never greater than 4 → we will never observe y_5
cardinality $M = 4$, alphabet $= \{y_1, y_2, y_3, y_4\}$

b) $P_Y(y = y_1) = P_X(x \leq -3) = \frac{1}{8}$

$P_Y(y = y_2) = P_X(-3 < x \leq 1) = 4 \cdot \frac{1}{8} = \frac{1}{2}$

$P_Y(y = y_3) = P_X(1 < x \leq 3) = 2 \cdot \frac{1}{8} = \frac{1}{4}$

$P_Y(y = y_4) = P_X(3 < x \leq 4) = \frac{1}{8}$

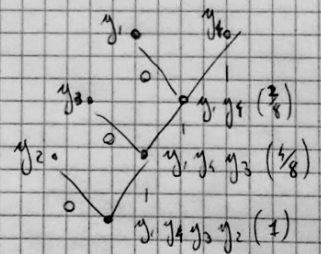
$P_Y(y = y_5) = P_X(x > 4) = 0$

c) $H(Y) = -\sum_{i=1}^{4} p(y = y_i) \log_2 p(y = y_i) = -\frac{1}{8}\log_2\frac{1}{8} - \frac{1}{2}\log_2\frac{1}{2} - \frac{1}{4}\log_2\frac{1}{4} - \frac{1}{8}\log_2\frac{1}{8} =$

$= 1{,}75$

d)

symbol	prob.	code
y_1	$\frac{1}{8}$	110
y_2	$\frac{1}{2}$	0
y_3	$\frac{1}{4}$	10
y_4	$\frac{1}{8}$	111



e) $L_H = \sum_{i=1}^{4} l_i^H \, p(y = y_i) = 3 \cdot \frac{1}{8} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} = 1{,}75$

$L_S = \sum_{i=1}^{4} l_i^S \, p(y = y_i) = \lceil -\log_2 \frac{1}{8} \rceil \cdot \frac{1}{8} + \ldots = 1{,}75$

NOTE: dyadic source → $H(Y) = L_H = L_S$

f) Given a p.d.f. and the extrema (a_i) of a quantizer interval, the optimal (in the MSE sense) reproduction value (y_i) is the centroid of the distribution within the interval
We are working with uniform distributions → centroid = mid-point

$y_1 = -3{,}5 \qquad y_2 = -1 \qquad y_3 = 2 \qquad y_4 = 3{,}5$

```
clc, clear all, close all

%% Parameters
N = 1000;
rho = 0.99;
w_std = 2;
clip = 15;
B = 2;

%% Signal generation (a)
w = randn(N, 1) * w_std;
x = zeros(N, 1);
x(1) = w(1);
for n = 2:N
    x(n) = rho*x(n-1) + w(n);
end

%% Signal generation (b)
x(x<-clip) = -clip;
x(x>clip) = clip;
x = round(x);

%% KLT encoding (a)
% reshape in blocks and remove mean
x_blocks = reshape(x, B, N/B);
x_mu = mean(x_blocks, 2);
x_blocks_zm = x_blocks - x_mu;

% compute autocorrelation matrix
RR = zeros(B, B, N/B);
for ii = 1:N/B
    RR(:, :, ii) = x_blocks_zm(:, ii)*x_blocks_zm(:, ii)';
end
RR_mean = mean(RR, 3);

% compute eigenvectors and eigenvalues of the autocorrelation matrix
[V, ~] = eig(RR_mean);

% define the transformation matrix
T_klt = V';

%% KLT encoding (b)
coeff = T_klt*x_blocks_zm;

%% KLT encoding (c)
coeff_q = round(coeff);

%% KLT decoding (a)
x_blocks_rec = T_klt'*coeff_q + x_mu;
x_rec = x_blocks_rec(:);

%% KLT decoding (b)
mse = mean((x - x_rec).*2);

%% Extra plot
figure(1), clf, hold on
plot(x)
plot(x_rec)
grid('On')
xlabel('samples')
ylabel('amplitude')
legend('x', 'x_{rec}')
title(sprintf('MSE = %.3f', mse))
```